

Experiment: 1

Date: 13-06-2026

Basic Linux Commands

AIM: To study the basic commands in Linux.

COMMANDS:

1.Task: To display the files and folders present in the login

Command: ls

Syntax: ls

Explanation: This command displays the files and folders present in the login.

2.Task: To view a file

Command: cat

Syntax: cat filename

Explanation: This command displays the contents of specified file.

3.Task: To create a file

Command: vi

Syntax: vi filename

Explanation: This command creates the file and content can be typed.

4.Task: To create a directory

Command: mkdir

Syntax: mkdir dirname

Explanation: This command creates a new directory with specified name.

5.Task: To change directory

Command: cd

Syntax: cd directory_name

Explanation: This command used to switch one directory to another.

6.Task: To come out of sub directory

Command: cd..

Syntax: cd..

Explanation: This command helps to switch to main directory.

7.Task: To display the current working directory

Command: pwd

Syntax: pwd

Explanation: This command displays current working directory showing the path.

8.Task: To delete a directory

Command: rmdir

Syntax: rmdir directory_name

Explanation: This command is used to delete the specified directory.

9.Task: To copy a file

Command: cp

Syntax: cp source_file destination_file

Explanation: This command produces a copy of the stored file and is stored.

10.Task: To remove or move a file

Command: mv

Syntax: mv source_file destination_file

Explanation: After moving content of source file to destination, then source file is deleted.

11.Task: To delete a file

Command: rm

Syntax: rm filename

Explanation: This command deletes the specified file from directory.

12.Task: To retrieve a part of file

Command: head

Syntax: head -no._of_rows filename

Explanation: This command displays no. of rows from the top of the specified file

13.Task: To retrieve a part of file

Command: tail

Syntax: tail -no._of_rows filename

Explanation: This command displays no. of rows from the bottom of specified file

14.Task: To sort the contents of a file

Command: sort

Syntax: sort filename

Explanation: This command sort the contents of a file in ascending order.

15.Task: To display the no. of characters in a file

Command: wc

Syntax: wc filename

Explanation: This command displays the no. of lines, words and characters of file

16.Task: To compare the contents of two files

Command: cmp

Syntax: cmp file1 file2

Explanation: This command compares a given file and display the area which it differs.

17.Task: To find the difference between 2 files

Command: diff

Syntax: diff file1 file2

Explanation: This command compares a given file and display the area at which differs.

18.Task: To match given pattern

Command: grep

Syntax: grep [-v|-n|-c] pattern filename

Explanation: This command verifies filename and checks whether pattern present in file or not.

19.Task: To give permissions to given file

Command: chmod

Syntax: chmod 777 filename

Explanation: This command gives the corresponding permission to user, group and others.

20.Task: To check display the history.

Command: history

Syntax: history

Explanation: This command display the history of showing path of directory and files.

21.Task: To fetch help

Command: help

Syntax: ls -help

Explanation: This command fetches to check the available directories.

Output:

Task 1,4:

```
[root@localhost ~]# mkdir Mitun
[root@localhost ~]# ls
Mitun      dos        hello.c    hello.js
[root@localhost ~]#
```

Task 2:

```
[root@localhost ~]# cat Skcet
My name is Mitun
I study in SKCET
Today is a good day
I love to watch movies
I like to play games
[root@localhost ~]#
```

Task 3:

```
My name is Mitun
I study in SKCET
Today is a good day
I love to watch movies
I like to play games
~
~
~
~
~
~
~
```

Task 5,6:

```
[root@localhost Mitun]# mkdir Mitun
[root@localhost Mitun]# ls
Mitun
[root@localhost Mitun]# cd Mitun
[root@localhost Mitun]# cd ..
[root@localhost Mitun]#
```

Task 7:

```
[root@localhost Mitun]# pwd
/root/Mitun
[root@localhost Mitun]#
```

Task 8:

```
[root@localhost Mitun]# rmdir Mitun
[root@localhost Mitun]# ls
[root@localhost Mitun]#
```

Task 9,10:

```
[root@localhost Mitun]# cp Skcet School
[root@localhost Mitun]# ls
Mitun  School  Skcet
[root@localhost Mitun]# mv School Study
[root@localhost Mitun]# ls
Mitun  Skcet  Study
[root@localhost Mitun]#
```

Task 11:

```
[root@localhost Mitun]# rm Study
[root@localhost Mitun]# cd ..
[root@localhost ~]# ls
Mitun  Skcet  dos      hello.c  hello.js
[root@localhost ~]#
```

Task 12,13:

```
[root@localhost ~]# head -2 Skcet
My name is Mitun
I study in SKCET
[root@localhost ~]# tail -3 Skcet
Today is a good day
I love to watch movies
I like to play games
[root@localhost ~]#
```

Task 14:

```
[root@localhost ~]# sort Skcet
I like to play games
I love to watch movies
I study in SKCET
My name is Mitun
Today is a good day
[root@localhost ~]#
```

Task 15:

```
[root@localhost ~]# wc School
      5      23      98 School
[root@localhost ~]#
```

Task 16,17:

```
[root@localhost ~]# cat Skcet
My name is Mitun
I study in SKCET
Today is a good day
I love to watch movies
I like to play games
[root@localhost ~]# cat skcet
I love to play mobile games
I like to play Badminton
I love to swim
I completed my schooling
[root@localhost ~]# cmp Skcet skcet
Skcet skcet differ: char 1, line 1
[root@localhost ~]# diff Skcet skcet
--- Skcet
+++ skcet
@@ -1,5 +1,4 @@
-My name is Mitun
-I study in SKCET
-Today is a good day
-I love to watch movies
-I like to play games
+I love to play mobile games
+I like to play Badminton
+I love to swim
+I completed my schooling
[root@localhost ~]#
```

Task 18:

```

[root@localhost ~]# grep -v play skcet
I love to swim
I completed my schooling
[root@localhost ~]# grep -c play skcet
2
[root@localhost ~]# grep -n play skcet
1:I love to play mobile games
2:I like to play Badminton
[root@localhost ~]#

```

Task 19:

```

[root@localhost ~]# chmod 777 skcet
[root@localhost ~]# ls
Mitun      School    Skcet     dos       hello.c   hello.js  skcet
[root@localhost ~]#

```

Task 20:

```

[root@localhost ~]# history
0 ls -l
1 qjs
2 qjscalc
3 qjsc
4 emacs hello.js
5 vi hello.js
6 ls
7 ls -l
8 qjs hello.js
9 qjsc hello.js
10 ls -l
11 qjsc hello.js -o hello
12 ls -l
13 qjsc
14 ls -l /usr/local/lib
15 ls -l /usr/lib/quickjs/libquickjs.*
16 strace qjsc hello.js
17 ls -l /tmp
18 ls /usr/include/quickjs/quickjs.h
19 gcc
20 strace qjsc hello.js -o hello
21 strace -f qjsc hello.js -o hello
22 ls
23 qjsc hello.js -o hello
24 ls -l
25 strace -f qjsc hello.js -o hello
26 ls

```


Task 21:

```
[root@localhost ~]# ls -help
ls: invalid option -- 'e'
BusyBox v1.31.0 (2019-09-15 13:46:40 CEST) multi-call binary.

Usage: ls [-lAaCxdLHRfpFli-n-s-lc-lu] [-w WIDTH] [FILE]...

List directory contents

    -l      One column output
    -a      Include entries which start with .
    -A      Like -a, but exclude . and ..
    -x      List by lines
    -d      List directory entries instead of contents
    -L      Follow symlinks
    -H      Follow symlinks on command line
    -R      Recurse
    -p      Append / to dir entries
    -F      Append indicator (one of */=@|) to entries
    -l      Long listing format
    -i      List inode numbers
    -n      List numeric UIDs and GIDs instead of names
    -s      List allocated blocks
    -lc     List ctime
    -lu     List atime
    --full-time  List full date and time
    -h      Human readable sizes (1K 243M 2G)
    --group-directories-first
    -S      Sort by size
    -X      Sort by extension
    -v      Sort by version
    -t      Sort by mtime
    -tc     Sort by ctime
    -tu     Sort by atime
    -r      Reverse sort order
    -w N    Format N columns wide
    --color[={always,never,auto}]  Control coloring
```